

Relation Algebra

§9. $\frac{R}{S}$

(9a)

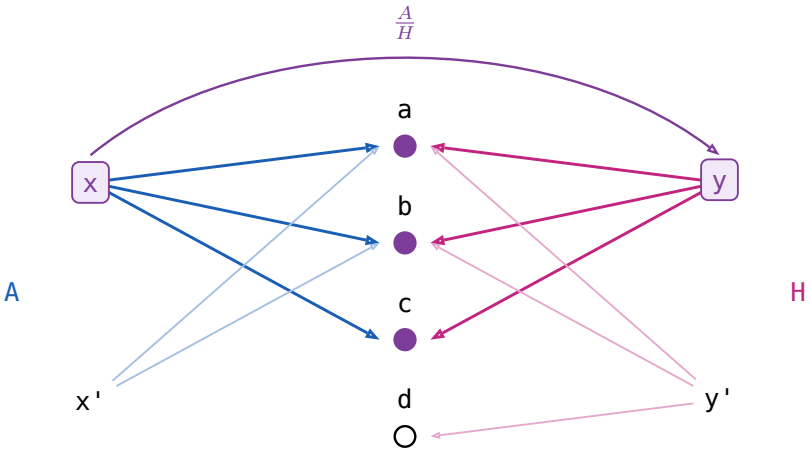
$R /_s S \triangleq (R / S) \cap (S / R)^\circ$ what R sends x to and nothing else is what S sends y to. In $\mathbf{Rel}$ x and y has the same image: $\forall p. (x R p \iff y S p)$

(9b)

- $x (A/H) y$ — x admires everyone y hates
- $x ((H/A)^\circ) y$ — x admires only people y hates
- $x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



	law	the reading
1	$X \sqsubseteq R /_s S \iff X S \sqsubseteq R \wedge X^\circ R \sqsubseteq S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S.	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
2	$(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.	Matching is symmetric.
3	$(R /_s S) (S /_s W) \sqsubseteq R /_s W$ a match from R to S followed by one from S to W is a match from R to W.	And transitive.
4	$(R /_s S) S \sqsubseteq R$ following the matching by S lands inside R.	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$
5	$(R /_s R) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people y=x always qualifies ($1 \sqsubseteq R\%R$ below)
6	$1 \sqsubseteq R /_s R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
7	$(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
8	$\text{Symmetric}(X) \implies X \sqsubseteq R /_s R \iff X R \sqsubseteq R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R.	The largest symmetric arrow that leaves R alone.
9	$\text{simplePart}(R) \triangleq R /_s 1$ matching R against the identity keeps the X whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1=R$.
10	$\text{dom}(R /_s S) = 1 \cap ((R / S) (S / R))$ the X that match something are those the two one-sided divisions take round the loop back to X.	Its domain is the domain of simplicity of R.